

EX No: 9

Date:

IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER SIMULATOR

AIM:

To implement subnetting in Cisco Packet Tracer Simulator and configure a network topology using subnetted IP addresses.

CREATE A NETWORK TOPOLOGY:

The first step in implementing classful IP subnetting is to create a network topology in Packet Tracer. Select the "New" option to create a new network. This will create a blank network topology where devices can be added.

SUBNETTING:

The subnet network address of 192.168.10.0/24 is subnetted to provide enough space for at least 5 addresses for each device/switch/router. We can use a /27 subnet mask. This gives us 8 subnets with 30 usable host addresses in each subnet.

The IP addressing for the network shown in the topology can be:

Device / Interface	IP Address
Router R – GigabitEthernet 0/0	192.168.1.1
Router R – GigabitEthernet 0/1	192.168.2.1
PC1	192.168.1.11
PC2	192.168.1.12
PC3	192.168.1.13
PC4	192.168.1.14
Fast Ethernet 0/1	192.168.3.0/27
PC1	192.168.4.11
PC2	192.168.4.12
PC3	192.168.4.13
PC4	192.168.4.14

Replace the IP address and subnet mask with your desired IP address and subnet mask.

Configure Terminal:

```
interface FastEthernet 0/1
switchport mode access
exit
interface FastEthernet 0/1
switchport mode access
exit
```

TESTING NETWORK:

Now our network topology is configured. Open a Command Prompt on each PC and try to ping the required PCs.

Last Status	Source	Destination	Type
Successful	PC4(12)	Router	ICMP
Successful	PC4(2)	PC2(10)	ICMP
Successful	PC0	Router	ICMP

Type	Color	Time	Num
ICMP	Green	0.000	12
ICMP	Blue	0.000	13
ICMP	Blue	0.000	14

RESULT:

The PCs connected to different switches can communicate with each other via routers.

The successful pings using the network properly configured show that the network is configured and operational.